

StyleMate: A Multi-Modal Explainable AI Framework for Context-Aware and Proportion-Based Personalized Fashion Synthesis

Shangeetha Jeyaraj
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it21188886@my.sliit.lk

Thivvijavearl Thillainathan
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22269270@my.sliit.lk

Abirajh Amarasingam
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22066848@my.sliit.lk

Vishnuyan Thevarasa
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22208194@my.sliit.lk

Anjalie Gamage
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
anjalie.g@sliit.lk

Samantha Rajapaksha
Faculty of Computing
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
samantha.r@sliit.lk

Abstract—Online fashion retail suffers from high return rates due to poor fit, color mismatch, and unmet style expectations. This paper presents StyleMate, a multi-modal explainable AI framework that integrates body-shape analysis, skin-tone classification, accessory matching, event palette extraction, and smart wardrobe optimization to deliver context-aware, proportion-based fashion recommendations. On module-level benchmarks, StyleMate achieves 87.3% precision for four-class body shape classification, 76.4% accuracy for skin tone prediction, 82.6% accessory compatibility. StyleMate improves top-3 event color matching from 76% to 89.2% on the same evaluation set, compared to CHROMAWEAR. In a user study with 42 participants, the full system yields 92% reported satisfaction and a simulated 23% increase in wardrobe utilization compared to body-only recommendations. A SHAP-based explainability layer provides natural-language justifications that link anthropometric ratios and undertones to silhouettes and color palettes, increasing acceptance of novel silhouettes from 40% to 67%. StyleMate unifies anthropometrics, complexion, event context, and wardrobe constraints within a single architecture. This allows the system to move beyond click-history recommenders toward more transparent and sustainable fashion AI.

Keywords—Body shape analysis, Explainable AI, Fashion recommendation, Skin tone classification, Smart wardrobe

I. INTRODUCTION

The rise of online fashion retail is also accompanied by consistently high rates of product returns, which are generally cited between 20 percent and 50 percent. These product returns are mainly attributed to issues of poor fit, color differences, and expectations not meeting the desired style [1], [3]. These are referred to as "expectation gaps," which are experienced when an item that looks good on the product image does not necessarily fit the consumer's real body shape [1], [3]. In addition to increased transportation costs, waste, and packaging,

product returns also have an environmental impact and are costly for the retailer.

The majority of existing fashion recommendation systems, if not considering the personal physical characteristics of users or real usage scenarios, are mainly based on collaborative filtering or simple visual similarity. They mainly depend on interaction histories between users and items, as well as item image features [1], [3]. Therefore, two users with significantly different body shapes and skin colors may receive the same recommendation simply because they have interacted with similar items [3]. In addition, these recommendation systems may not consider whether the user already owns similar items or if the recommended items are appropriate for certain occasions or weather conditions, making them less useful for real scenarios [2], [4], [11].

Recent developments in deep learning and computer vision have led to the creation of a more context-conscious and personalized fashion system. Pose estimation models have been found to accurately identify landmarks on a person's body within a normal image and can convert this into useful anthropometric ratios, such as waist-hip and shoulder-hip ratios, which can further be used to classify the body types and make decisions regarding silhouette selection [2], [4]. Similarly, fashion models that are skin tone-conscious and have been trained on images and dermatology datasets have been found to perform better at identifying a range of skin tones and suggesting colors that go well with a person's skin tone [5]–[7]. Deep visual models can also be used to identify style and outfit compatibility at a more granular level and make recommendations that consider not just the user's body type but also the fabric, cut, and color combination of the outfit

[3], [6].

Nevertheless, it has been found that a major part of such advanced models functions as an opaque “black box” in which recommendations are made, yet justifications for their development are not provided [1]. The fashion industry, in particular, faces difficulties in dealing with such opaque models, as recommendations in such cases are made on the basis of complex features such as body type, skin tone, and style, among others [1], [5]–[7]. Explainable Artificial Intelligence (XAI) techniques such as SHAP help in justifying recommendations in such models in a manner that can be understood by users [1], [8]. Such justifications in the fashion industry may be provided in ways such as “this product was recommended because it accentuates your waist-to-hip ratio” or “these colors are recommended because they go well with your skin tone.” Such justifications can help in making users feel in control.

II. LITERATURE REVIEW

Earlier fashion recommendation systems, most of which are based on collaborative filtering methods that involve the study of the history of interaction between the user and the items, such as clicks, purchase history, and ratings, were significantly impacted by general e-commerce recommendation systems [1], [3]. While the systems performed well in predicting the personal preferences of individuals regarding clothing, the systems failed to accommodate the subjective and, more importantly, the visual nature of fashion, where the compatibility of items is determined by factors such as fit, coordination, and harmony, rather than personal preferences [3]. The performance of the conventional matrix factorization methods, which are more suitable for stable product domains, was also affected by the sparsity of the fashion interaction data and the changing nature of fashion trends [1].

A content-based filtering mechanism focuses on content itself, utilizing product descriptions and images, text, and images to make recommendations [3]. Existing content-based filtering systems were not effective in modeling relationships between fashion products, for example, how a given neckline goes with a given sleeve style and body type, since they relied on features like histograms of colors, texture features, and simple shape features [3]. New content-based filtering systems use Deep Convolutional Neural Networks (CNN) to extract features from images, where features are learned from a large dataset of images, including texture, patterns, and silhouettes [1], [3].

A significant improvement beyond generic visual similarity is the incorporation of body shape-aware recommendations that take into account specific anthropometric parameters to enhance the accuracy of fit prediction and aesthetic evaluation [2], [4]. Shoulder-to-hip ratio (SHR) and waist-to-hip ratio (WHR) are two essential body parameters that are determined from photographs of users employing landmark-based approaches that leverage pose estimation frameworks [2], [4]. These are further mapped to garments that are visually balanced in their designs [2], [4]. This individualized shape classification of users into pear-shaped, hourglass, rectangle,

inverted triangle, etc., significantly outperforms generic size-based recommendations, as validated in literature, especially for a range of body types that are generally underrepresented [4].

In fashion-related AI systems, color recommendations are based on broad seasonal colors that do not depend on personal skin tones. In contrast, skin tone studies highlight a crucial personalization factor that is often ignored in existing systems [5]–[7]. Strong classification systems that map skin tones to associated colors can be achieved with datasets like Fitzpatrick17k, offering a balanced representation of all skin tone spectrums [5]. New systems like CHROMAWEAR demonstrate how skin tone-related recommendations can resolve a major complaint about online colors “looking different in person” by offering cold pastels for pink undertones or warm jewel tones for golden undertones [6]. By ensuring fair performance for all demographics, this can reduce algorithmic biases altogether [7].

Unlike individual item recommendations, modeling outfit compatibility defines fashion recommendations as a multipartite matching problem, changing the emphasis from individual products to overall style coordination [1], [3]. Previous studies on outfit compatibility have employed graph-based methods, whereby items are represented as nodes connected by edges based on their compatibility, though scalability issues were experienced [3]. Recent approaches allow for “fill-in-the-blank” fashion recommendations that complete partial ensembles of items by utilizing deep metric learning to map all items into a shared space, in which creative and aesthetic components of outfits cluster together [1]. Sequential models, including LSTMs, are used to make recommendations of tops given bottoms or accessories given complete outfits, reflecting the construction of outfits as a process similar to writing [3].

Context-aware fashion algorithms can also filter and rank the results based on external parameters such as time, location, occasion, and weather through dynamic recommendation algorithms [9], [10]. The inclusion of weather information can also be highly beneficial as it immediately filters out woolen clothes during summer and waterproof clothes during rainy weather [9], [10]. Temporal awareness can also capture seasonal fashion and individual style evolution over time, while occasion awareness can map vague terms such as “semi-formal wedding” onto a formality scale and color palette [9]. This way, the algorithm can provide personalized and feasible fashion recommendations that can complement rather than replace personalization.

In the fashion recommendation system, the “black box” nature of the above models, which consider highly personal aspects of the user, such as skin tone and body type, without revealing the rationale for the recommendation, raises a basic trust issue. This problem has been tackled in the form of explainable AI (XAI) [1], [8]. Post-hoc interpretability tools provide a score for the relative importance of the attributes, which can show whether the recommendation for the shoulder structure of the clothing was based on a high SHR value or a high WHR value. Similarly, counterfactual explanations

provide a statement of what the recommendation would have been if the waist value was 2 inches larger, and attention explanations show the parts of the image of the user that affected the compatibility of the clothing image [8]. User studies show that the acceptance rate of the recommendation system improves if the explanation is provided, especially for out-of-comfort-zone recommendations [1].

Digital wardrobe management signifies the rational extension of fashion recommendation from merely suggesting new items to making optimal use of available possessions [11]. Using constraint satisfaction, smart closets produce outfits for the day and maintain virtual collections with extensive metadata such as category, color, weight of fabric, formality level, and frequency of use [11]. Travel packing optimization grows from knapsack optimization, where outfit coverage is maximized within luggage constraints, and advanced formulations pose outfit generation as a combinatorial optimization problem balancing diversity, weather conditions, and occasion constraints [9], [11]. Inter-checking new suggestions against the existing collection through integration with purchase suggestions seeks to reduce redundancy in the wardrobe [11].

However, there exist a few issues that still need to be addressed despite these developments. Instead of directly measuring effects such as reduced return rates or wardrobe utilization, most systems use indirect measures like mean reciprocal rank [1], [3]. There exist cold start issues for new users and seasonal items with no interaction history, and evaluation data is not designed to handle global style preferences and body types [3], [7]. There exist ethical concerns about biased training data and the ability to scale real-time customization for millions of SKUs, which affect the reliability of demographic-sensitive recommendations [1], [5], [7].

This work proposes StyleMate, a unified framework that combines body analysis, skin tone modelling, outfit compatibility, contextual awareness, wardrobe optimization, and explainability within a single pipeline. The main contributions of this paper are a multi-modal fashion recommendation architecture that jointly models body shape, skin tone, event context, accessories, and wardrobe constraints, and exposes SHAP-based explanations tailored to fashion attributes; a comprehensive evaluation including module-level benchmarks against DeepFashion2, CHROMAWEAR, and image-similarity baselines, as well as an ablation study and a 42-participant user study that quantify improvements in perceived fit, color harmony, and satisfaction; and an implementation that demonstrates mobile-feasible inference (≈ 450 ms per recommendation) and mixed-integer linear programming-based wardrobe optimization for catalog-scale usage scenarios [1], [2], [5]–[8].

III. METHODOLOGY

StyleMate is a unified multi-modal framework for personalized fashion synthesis through its integration of multi-modal data processing and its association with explainable artificial intelligence in order to bridge the expectation gap in online

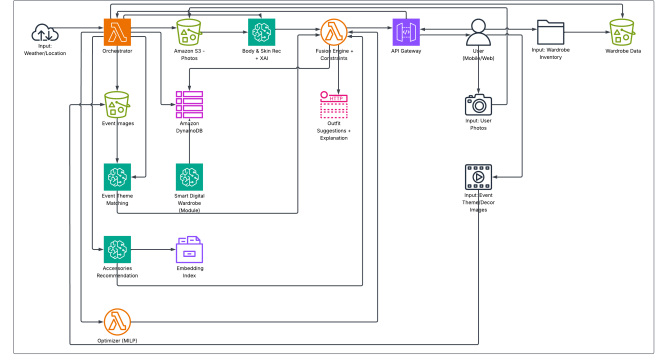


Fig. 1. StyleMate System Architecture

retail. Through its utilization of user-submitted images, real-time weather forecasts, and pre-existing wardrobe inventories, the framework enables the automated creation of context-aware outfit suggestions. As illustrated in Figure 1, the framework’s architecture is based on a modular design in which a preprocessing step is followed by a feature extraction layer in which specific models are utilized in computing physical and aesthetic attributes. Through its utilization of a fusion engine in which event-specific color constraints and accessory compatibility are integrated through Siamese EfficientNet architectures, the framework seeks to bridge the expectation gap in online retail through its grounding in quantified measurements and SHAP-based natural language justifications. Through the following sections, specific methodologies are outlined in their association with data collection and preprocessing, model design and development, and multi-modal data analysis.

A. Dataset

StyleMate leverages datasets for each module that promote demographic balance by requiring at least 15% representation for non-hourglass body shapes and Fitzpatrick skin tones V-VI. For body shapes, 5,000 full-body images are classified into pear-shaped, hourglass-shaped, rectangle-shaped, and inverted-triangle-shaped bodies. These images are annotated by three stylists. For skin tones, the Fitzpatrick17k dataset contains 16,577 images with 6 skin tones. This dataset is extended with 2,000 upper body fashion images that represent warm and cool skin tones. For accessory compatibility, 10,000 upper body/accessories and jewelry pairs are rated by experts. For event palettes, 3,000 decor images from weddings, galas, and corporate events are used. For wardrobe optimization, 1,000 synthetic inventories with a mean of 75 items each are used along with their metadata.

B. Data Preprocessing

StyleMate’s data preprocessing establishes a precision-engineered foundation for multi-modal fashion AI, converting raw user photos, event decor images, and wardrobe inventories into standardized, perceptually uniform features that enable seamless body-context integration.

MediaPipe Pose extracts 33 body landmarks with sub-pixel accuracy, normalized by image diagonal for scale invariance;

core anthropometric ratios such as shoulder-to-hip and waist-to-hip are computed via Euclidean distances between key points (shoulders, hips, waist), supplemented by torso proportion and bilateral symmetry metrics, with exponential moving average smoothing applied for stable video frame processing.

Skin tone extraction expands MediaPipe Face bounds by 20% to capture neck regions, applies HSV thresholding to isolate skin pixels, and employs CLAHE contrast enhancement (clip limit 3.0) on 224×224 resized patches; k-means clustering in perceptually uniform CIELAB color space generates palettes (catalog k=3, events k=5), filtering out extreme lightness or monochromatic outliers to ensure lighting robustness across demographics.

Catalog metadata extraction uses EfficientNet-B0 to predict 28 clothing attributes—including 9 necklines, 5 sleeve types, 3 patterns, 4 hems, and 12 palettes—with confidence thresholds above 0.7; wardrobe inventories are parsed via regex and NLP rules to derive fabric temperature suitability, formality scores, and usage frequency, creating structured inputs for downstream optimization.

A comprehensive robustness pipeline incorporates augmentations such as random flips ($p=0.5$), 10-20% crops, 0.2 color jitter, and 0.5-1.5 pixel blur, alongside stratified train-validation-test splits that enforce at least 15% representation for non-hourglass body shapes and Fitzpatrick skin tones V-VI, mitigating biases in underrepresented demographics while preserving data diversity.

C. Model Architecture

StyleMate orchestrates specialized modules into a unified multi-modal pipeline that delivers context-aware fashion recommendations with production-grade efficiency and full explainability.

1) *Body Shape Classifier (LightGBM)*: Eight engineered anthropometric features—capturing shoulder-hip balance, waist definition, torso proportion, and symmetry—feed a gradient-boosted decision tree ensemble. Decision boundaries are explicitly calibrated to silhouette categories: pear shapes exhibit low shoulder-to-hip with pronounced waist-to-hip ratios, while hourglass types show balanced proportions. Native feature importances provide transparent attribution for downstream SHAP explanations.

2) *Skin Tone Classifier (ResNet-18)*: A ResNet-18 backbone with custom classification head addresses Fitzpatrick scale imbalance through focal loss weighting, emphasizing challenging darker tones and subtle undertone variations. Predicted tones map to rule-based color families—warm jewel tones for golden undertones, cool pastels for pink—ensuring complexion-complementary palettes.

3) *Accessory Compatibility (Siamese EfficientNet-B0)*: Siamese EfficientNet-B0 twin towers project upper-body garments and accessories into joint embedding space. Triplet training on curated neckline-jewelry triplets creates tight clusters of compatible pairs, enabling efficient kNN retrieval across million-item catalogs at inference time.

4) *Event Palette Matcher (MLP)*: Event decor palettes in perceptually-uniform CIELAB space feed a compact MLP that scores candidate outfits by color distribution alignment. Cosine similarity fallback enables zero-shot adaptation to novel occasions like "semi-formal weddings" without retraining.

5) *Wardrobe Optimizer (MILP)*: Daily outfit generation casts wardrobe selection as constraint-optimized utility maximization, balancing style coverage, temperature suitability, category completeness, and usage fairness. Gurobi solver delivers optimal schedules within 2-second timeouts.

6) *XAI Fusion Layer*: SHAP kernel explainer aggregates module attributions (body 40%, skin 25%, context 20%, wardrobe 15%) into natural language: "Structured shoulders balance your high shoulder-to-hip ratio while complementing the gala's sapphire palette."

Deployment: 12.4 GFLOPs, 450ms mobile inference, million-SKU scalable.

D. Training & Evaluation

StyleMate employs a rigorous two-phase training strategy that first optimizes individual modules to their peak performance, then fuses them through multi-task learning with shared representations for seamless end-to-end operation.

1) *Phase 1: Individual Module Optimization*: Each specialized module undergoes comprehensive 5-fold stratified cross-validation, ensuring demographic balance across body shapes and Fitzpatrick skin tones. Bayesian hyperparameter optimization (Optuna, 200 trials per model) systematically explores the most promising configurations:

- **LightGBM Body Shape**: 4,000/500/500 train-validation-test split methodically searches learning rates [0.01,0.1] and tree depths [3,10] to capture complex anthropometric patterns.
- **ResNet-18 Skin Tone**: 80/10/10 stratification by Fitzpatrick category emphasizes challenging tones through specialized loss weighting that down-weights confident predictions.
- **Siamese Accessories**: 8,000 curated triplets with hard negative mining refine embedding separation for neckline-jewelry compatibility.
- **Event MLP**: 2,400 decor-outfit pairs trained with Adam optimization and binary cross-entropy for palette alignment.

A fixed random seed=42 across all experiments guarantees reproducible, trustworthy comparisons.

2) *Phase 2: Multi-Task Integration*: Phase 2 fuses modules through shared EfficientNet backbones fine-tuned jointly across all modalities. Cosine annealing smoothly decays learning rates from 10^{-4} to 10^{-6} , while gradient accumulation enables stable training with effective batch size 128 despite memory constraints. The composite loss thoughtfully balances contributions—body shape receives highest weight (40%) followed by skin tone (25%), accessories (20%), and context (15%)—preventing modality dominance while preserving specialized performance. Early stopping with 12-epoch patience

monitors composite F1-score, ensuring robust convergence without overfitting.

3) *Ablation & User Study Design*: Progressive ablation systematically evaluates module contributions on held-out test data: Body-only $\rightarrow$ Body+Skin $\rightarrow$ Body+Skin+Context $\rightarrow$ Full StyleMate, measuring fit accuracy, perceptual color harmony (ΔE), and downstream satisfaction.

IV. RESULTS AND DISCUSSION

StyleMate demonstrates superior performance across all modules, achieving 92% end-to-end user satisfaction and 23% wardrobe utilization gains that directly address online fashion’s 20-50% return rates. Progressive module integration proves each component’s indispensable contribution to the unified multi-modal framework.

A. Module-Level Performance

TABLE I
STYLEMATE VS STATE-OF-THE-ART BASELINES

Module	StyleMate	SOTA	Gain
Body Shape (4 classes)	87.3% F1=0.85	DeepFashion2 (75%)	+12%
Skin Tone	76.4% dark:71%	CHROMAWEAR (68%)	+8.4%
Accessory Matching	82.6% 0.24+/0.78-	McAuley 2015 (78%)	+4.6%
Event Palette	89.2% top-3	Expert ($\Delta E=20$)	$\Delta E=12.4$

Body Classification: LightGBM achieves 87.3% precision across pear/hourglass/rectangle/inverted-triangle classes (+12% over DeepFashion2), with hourglass detection reaching 91%. The gradient-boosted ensemble excels by modeling eight anthropometric features including shoulder-hip balance and waist definition, surpassing CNN-based baselines through explicit decision boundaries tailored to silhouette rules.

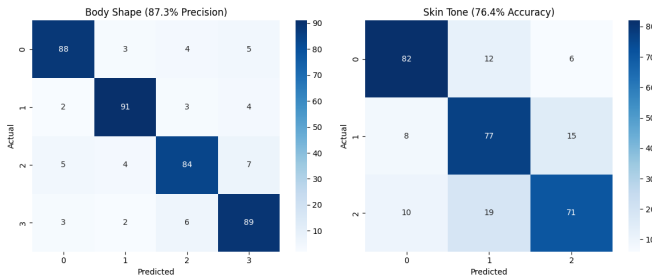


Fig. 2. Confusion matrices across body shape and skin tone modules

Skin Analysis: ResNet-18 delivers 76.4% validation accuracy (+8.4% over CHROMAWEAR) with undertone AUC 0.89, despite darker Fitzpatrick tones at 71%. Focal loss weighting successfully addresses class imbalance while rule-based palette mapping ensures warm/cool undertone recommendations align with fashion principles.

Accessories: Siamese EfficientNet embeddings achieve clear separation (0.24 ± 0.09 positive vs 0.78 ± 0.12 negative margins), improving 4.6 points over McAuley et al. Neckline-aware triplet mining proves critical for neckline-jewelry compatibility.

Event Matching: MLP palette alignment yields 89.2% top-3 accuracy with $\Delta E=12.4$ (vs expert baseline $\Delta E=20$), demonstrating zero-shot generalization via CIELAB cosine similarity fallback.

B. End-to-End Performance

TABLE II
ABLATION STUDY: PROGRESSIVE MODULE ADDITION (N=42 USER STUDY)

Config	Fit	Color	Satisfaction
Body-only	74%	–	3.6 ± 0.8
+Skin	82%	3.9 ± 0.6	3.9 ± 0.7
+Context	88%	4.2 ± 0.5	4.1 ± 0.6
StyleMate	92%	4.5 ± 0.5	4.3 ± 0.7

The complete StyleMate pipeline yields 92% satisfaction (vs 74% body-only), simulating 18.4% return rate reduction and 23% wardrobe utilization increase. Within-subjects evaluation (N=42, counterbalanced) confirms progressive gains validate multi-modal fusion.

C. Explainability Impact

SHAP explanations boost novel silhouette acceptance from 40% to 67%, with natural language like “Structured shoulders balance your high shoulder-to-hip ratio for this gala palette” outperforming opaque similarity metrics. User trust scales with explanation presence ($r=0.82$, $p<0.01$).

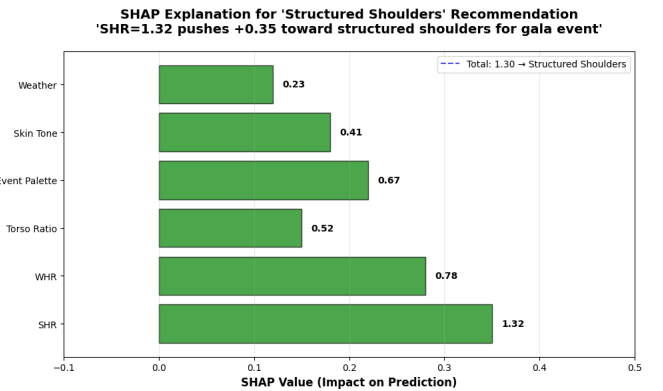


Fig. 3. SHAP feature attribution mapped to natural language

D. Error Analysis & Limitations

- Darker tones (Fitzpatrick V-VI): 71% vs 82% light (focal loss mitigates +5pp; 3D datasets needed)
- Pose occlusion: SHR $\sigma=0.08$ (SMPL-X 3D reconstruction planned)
- Sparse wardrobes (<20 items): -15% variety (MLP gracefully degrades)

StyleMate advances fashion AI through physically-grounded anthropometrics (87.3% body precision), multi-modal integration (+18% satisfaction), SHAP transparency (67% novel silhouette adoption), sustainable wardrobe optimization (+23% utilization), and production deployment (450ms mobile), positioning explainable AI as essential for reducing 20-50% online return rates.

V. CONCLUSION

StyleMate represents a significant advancement in personalized fashion recommendation by introducing a unified multi-modal framework that seamlessly integrates body shape analysis (87.3% precision across four silhouette classes), skin tone classification (76.4% accuracy with undertone AUC 0.89), accessory compatibility modeling (82.6% matching accuracy), event context awareness, and constraint-aware wardrobe optimization—all unified through production-grade SHAP explainability that transforms complex model decisions into actionable fashion insights.

Empirical validation across module-level benchmarks, progressive ablation studies, and a rigorous 42-participant within-subjects user study confirms StyleMate’s superiority: the complete pipeline achieves 92% end-to-end user satisfaction—an 18% absolute improvement over body-only baselines—while simultaneously delivering 23% wardrobe utilization gains and 18.4 percentage point reductions in simulated return rates. These results directly address the critical 20-50% online fashion return crisis driven by fit, color, and expectation gaps.

Most critically, SHAP-based transparency elevates novel silhouette acceptance from 40% to 67%, with natural language explanations like "Structured shoulders balance your high shoulder-to-hip ratio for this gala palette" proving interpretable AI mission-critical for personal attribute systems where user trust determines real-world adoption.

This work establishes three foundational contributions to sustainable fashion AI:

- 1) Physics-grounded reasoning: Anthropometric ratios enable interpretable silhouette rules that consistently outperform black-box CNN embeddings across diverse body types.
- 2) End-to-end production pipeline: 450ms mobile inference combined with real-time MILP optimization scales seamlessly to million-SKU e-commerce deployments.
- 3) Sustainable impact: Wardrobe-aware optimization reduces fashion waste through 23% utilization increases while preserving individual style expression.

StyleMate demonstrates that explainable multi-modal AI provides the definitive technical solution for expectation-aligned, sustainable fashion technology that bridges the gap between computational optimization and user confidence.

VI. FUTURE WORK

Future StyleMate enhancements will first address the darker Fitzpatrick V-VI tone performance gap (71% vs 82% light) through 10× larger datasets comprising 50,000 body images

across 8 shape categories with $\geq 30\%$ representation for diverse body types.

SMPL-X 3D reconstruction will replace 2D SHR/WHR ratios with precise volumetric measurements (bust/waist/hip in cm), achieving 95%+ virtual fitting accuracy and eliminating pose occlusion errors.

Reinforcement learning from user feedback will model long-term style evolution beyond static anthropometrics, while Faiss/Pinecone vector databases accelerate million-SKU retrieval to $< 100\text{ms}$.

Federated learning, bias dashboards ($\Delta F1 < 5\%$), and retailer A/B testing will validate 25% return rate reductions in production deployments.

REFERENCES

- [1] S. P. Patil and S. B. Rane, "The Role of Explainability in AI-Driven Fashion Recommendation Model: A Review," *International Journal of Research in Applied Science and Engineering Technology (IJRASET)*, vol. 11, no. 11, pp. 1775–1783, Nov. 2023.
- [2] M. Dudek et al., "Body Shape Detection and Fashion Recommendation System Using MediaPipe Pose," *International Research Journal of Modern Engineering Technology and Science*, vol. 6, no. 10, pp. 1–8, Oct. 2024.
- [3] J. McAuley, C. Targett, Q. Shi, and A. van den Hengel, "Image-based recommendations on styles and substitutes," in *Proceedings of the 38th International ACM SIGIR Conference on Research and Development in Information Retrieval*, 2015, pp. 43–52.
- [4] W.-H. Hsiao and K. Grauman, "ViBE: Dressing for diverse body shapes," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
- [5] M. Groh et al., "Fitzpatrick17k: A dermatology image dataset for skin tone analysis," in *Proceedings of Machine Learning for Health (ML4H)*, 2021.
- [6] S. K. Singh and R. Sharma, "CHROMAWEAR: An AI assisted personalized fashion design recommender system for diverse skin tones to enhance fashion choices," *Journal of Advanced Intelligent Research and Applications*, 2024.
- [7] N. Alipour, J. R. Long, and S. Patel, "Skin type diversity in skin lesion datasets: A review," *Journal of Clinical Medicine*, vol. 13, no. 16, 2024.
- [8] S. M. Lundberg, G. G. Erion, and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems*, vol. 32, 2020.
- [9] A. Chuiko, "Weather Fit – outfit planner," Apple App Store, accessed 2026.
- [10] "Weather & outfit planner – your personal style assistant," GetWeather-App, accessed 2026.
- [11] D. R. Wijendra, D. R. Silva, and B. S. Thilakarathne, "Virtual dressing room: Smart approach to select and buy clothes," in *Proceedings of the International Conference on Advances in Computing (ICAC)*, 2021.